Set 19: Survey Responses Analysis

DSA – Data Handling and Visualization | Lab Experiment

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Aim

To analyse survey answers using a grouped bar chart, a stacked bar chart and a labelled table, and to combine them in an interactive Tableau dashboard.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
library(patchwork)
options(scipen = 999)
# Plain white table theme with black borders (no colours)
tbl_theme <- ttheme_minimal(
  core    = list(fg_params = list(col = "black", fontsize = 11),
                 bg_params = list(fill = "white", col = "black")),
  colhead = list(fg_params = list(col = "black", fontface = "bold", fontsize = 11.5),
                 bg_params = list(fill = "white", col = "black")))
survey <- data.frame(
  Survey_ID  = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)
write.csv(survey, "Survey_Responses.csv", row.names = FALSE)  # for Tableau
survey_long <- data.frame(
  Question = rep(c("Question 1", "Question 2", "Question 3"), each = 3),
  Answer   = factor(c(survey$Question_1, survey$Question_2, survey$Question_3), levels =
c("A", "B", "C", "D")))
counts <- as.data.frame(table(Question = survey_long$Question, Answer = survey_long$Answer))
names(counts)[3] <- "Responses"
pal <- c(A = "#0072B2", B = "#E69F00", C = "#009E73", D = "#CC79A7")
```

Q1. Grouped Bar Chart – Distribution of Answers for Question 1

> *Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.*

R Program:

```
cat("Answer distribution for Question 1:\n")
print(table(Answer = factor(survey$Question_1, levels = c("A", "B", "C", "D"))))
q1 <- counts[counts$Question == "Question 1", ]
p_grp <- ggplot(q1, aes(x = Answer, y = Responses, fill = Answer)) +
  geom_col(width = 0.6, position = position_dodge(width = 0.8)) +
  geom_text(aes(label = Responses), vjust = -0.4, size = 4.5) +
  scale_fill_manual(values = pal) + scale_y_continuous(limits = c(0, 2), breaks = 0:2) +
  labs(title = "Distribution of Answers for Question 1", x = "Answer Option", y = "Number of
Responses", fill = "Answer") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_grp)
ggsave("s19_q1.png", p_grp, width = 7, height = 4.5, dpi = 150, bg = "white")
# Grouped version comparing all three questions
p_grp_all <- ggplot(counts, aes(x = Answer, y = Responses, fill = Question, group = Question))
+
  geom_col(position = position_dodge(width = 0.8), width = 0.75, colour = "grey30") +
  geom_text(aes(label = Responses), position = position_dodge(width = 0.8), vjust = -0.4, size
= 3.6) +
  scale_fill_manual(values = c("Question 1" = "#0072B2", "Question 2" = "#E69F00", "Question
3" = "#009E73")) +
  scale_y_continuous(limits = c(0, 2.5), breaks = 0:2) +
  labs(title = "Grouped Bar Chart: Answers by Question", x = "Answer Option", y = "Number of
Responses", fill = "Question") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
ggsave("s19_q1b.png", p_grp_all, width = 7.5, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
Answer distribution for Question 1:
Answer
A B C D
1 1 1 0
```

Output:

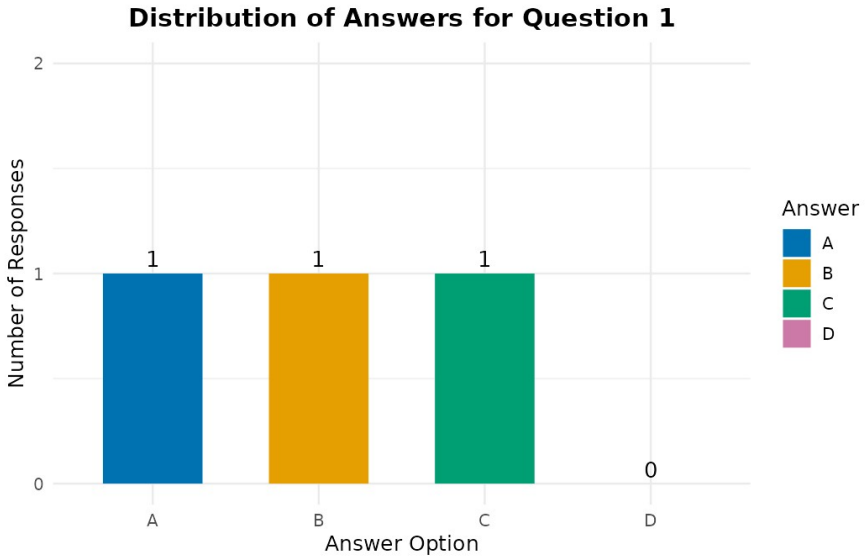


Figure 1: Distribution of answers for Question 1

Output:

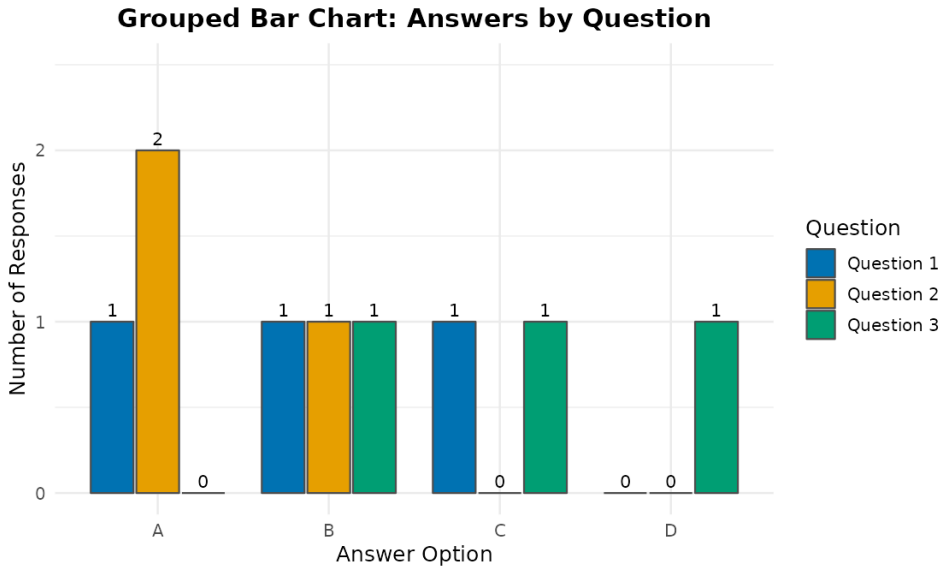


Figure 2: Grouped bars comparing the answer options across all questions (Question 1 in blue)

Inference:

For **Question 1**, answers A, B and C were each chosen once and D was never chosen, so responses are evenly split with no consensus. In the grouped chart, Question 2 has a clear majority for A (2 of 3).

Q2. Stacked Bar Chart – Overall Distribution for All Questions

> *Generate a stacked bar chart to represent the overall distribution of responses for all three questions.*

R Program:

```
cat("Answer counts for all three questions:\n")
print(xtabs(Responses ~ Question + Answer, data = counts))
cat("\nOverall totals:\n"); print(table(survey_long$Answer))
p_stk <- ggplot(counts[counts$Responses > 0, ], aes(x = Question, y = Responses, fill =
Answer)) +
  geom_col(width = 0.55, colour = "white") +
  geom_text(aes(label = Answer), position = position_stack(vjust = 0.5), colour = "white",
fontface = "bold", size = 4.5) +
  scale_fill_manual(values = pal) +
  labs(title = "Overall Distribution of Responses for All Questions", x = "Question", y =
"Number of Responses", fill = "Answer") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_stk)
ggsave("s19_q2.png", p_stk, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
Answer counts for all three questions:
            Answer
Question     A B C D
  Question 1 1 1 1 0
  Question 2 2 1 0 0
  Question 3 0 1 1 1

Overall totals:

A B C D
3 3 2 1
```

Output:

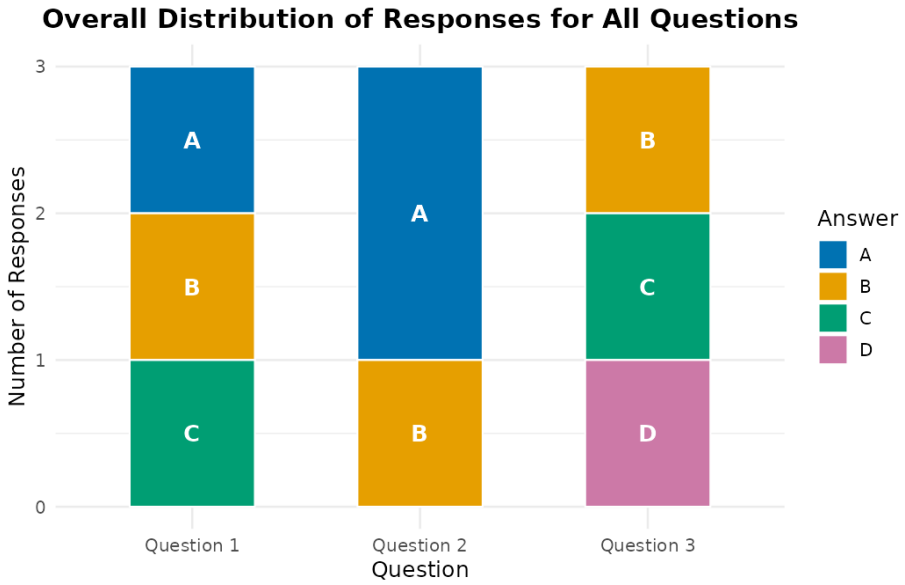


Figure 3: Stacked bar chart of responses per question

Inference:

Overall, A and B are the most common answers (3 each), C is next (2) and D is least common (1, only in Question 3). Question 2 is the most one-sided (A = 2), while Questions 1 and 3 spread their answers over three options.

Q3. Table – Survey Response Data

> *Build a table to show the survey response data for each survey. Label the table elements.*

R Program:

```
tbl <- survey
names(tbl) <- c("Survey ID", "Question 1", "Question 2", "Question 3")
print(tbl, row.names = FALSE)
tg <- tableGrob(tbl, rows = NULL, theme = tbl_theme)
png("s19_q3.png", width = 900, height = 260, res = 150)
grid.arrange(grid::textGrob("Survey Response Data", gp = grid::gpar(fontsize = 14, fontface =
"bold")),
             tg, ncol = 1, heights = c(0.3, 1))
invisible(dev.off())
```

Console Output:

```
 Survey ID Question 1 Question 2 Question 3
         1          A          B          C
         2          B          A          D
         3          C          A          B
```

Table Output:

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Inference:

The table shows each respondent's complete answer set. No two surveys gave the same answer pattern.

Q4. Tableau Dashboard – Interactive Exploration

> *Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.*

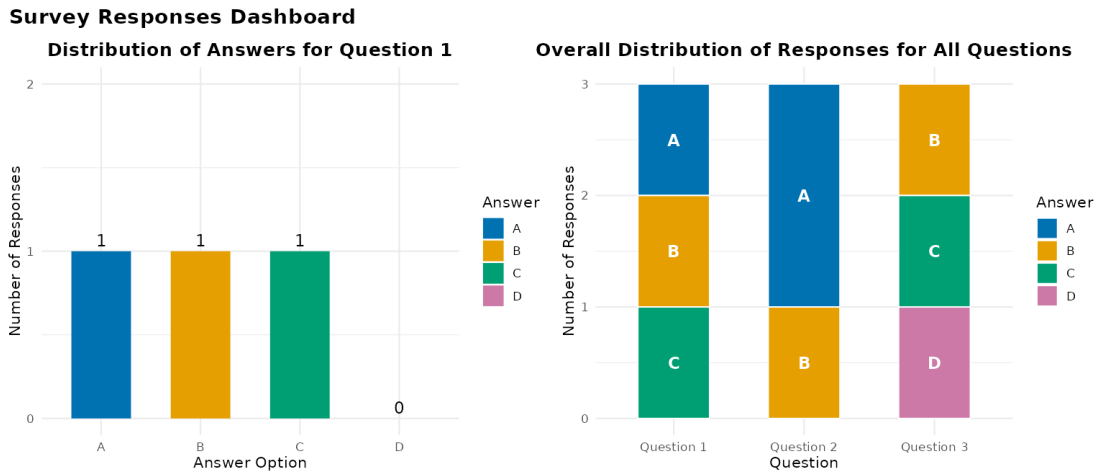
R Program:

```
dash <- (p_grp | p_stk) / wrap_elements(full = tg) + plot_layout(heights = c(1.4, 0.6)) +
  plot_annotation(title = "Survey Responses Dashboard",
                  theme = theme(plot.title = element_text(face = "bold", size = 16)))
ggsave("s19_dash.png", dash, width = 12, height = 7, dpi = 130, bg = "white")
```

Procedure in Tableau:

1. Open Tableau → **Connect → Text file** → **Survey_Responses.csv**. In the Data Source tab, select Question_1, Question_2 and Question_3 → right-click → **Pivot**. Rename the pivot fields to **Question** and **Answer**.
2. **Sheet 1 – "Grouped Bar":** drag Answer to Columns, Question to Columns (right of Answer) and **CNT(Survey ID)** to Rows. Drag Question to Color and show labels. Add a Question filter set to **Question 1** to focus on it.
3. **Sheet 2 – "Stacked Bar":** drag Question to Columns and CNT(Survey ID) to Rows, then drag Answer to **Color** and Label.
4. **Sheet 3 – "Response Table":** drag Survey ID and Question to Rows and Answer to Text (or use the original un-pivoted columns as a crosstab). Rename the headers.
5. From the menu choose **Dashboard → New Dashboard** and set **Size → Automatic** (or Fixed 1200 × 800).
6. Drag the sheets onto the canvas: Grouped Bar (top-left), Stacked Bar (top-right), Response Table (bottom). Add a **Text** object at the top as the dashboard title.
7. On each chart, open the sheet's drop-down and select **Use as Filter**, so clicking a bar, point or slice filters the other views.
8. Go to **Dashboard → Actions → Add Action → Highlight** (Run on: Hover) so related marks highlight across all views.
9. Right-click each quick filter → **Apply to Worksheets → All Using This Data Source** (Survey_Responses) so one filter controls every view.
10. Save as a packaged workbook (**File → Save As → .twbx**) or publish to Tableau Public.

Dashboard Layout:



Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Reference layout of the dashboard (drawn in R). The interactive version is built in Tableau using the procedure above.

Inference:

Clicking an answer segment (for example, A) in the stacked bar highlights the surveys in the table that gave that answer, so each response can be traced back to its respondent.

Result

The survey responses were visualised with grouped and stacked bar charts and a labelled table in R, and combined into an interactive Tableau dashboard (reshaped with a pivot). Question 1 is evenly split across A, B and C; Question 2 favours A; and A and B are the most frequent answers overall.